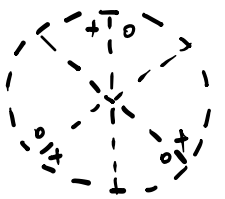
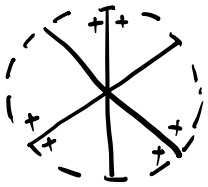
revisit :

一.



D_3

D_3 和 C_{3v}



C_{3v}

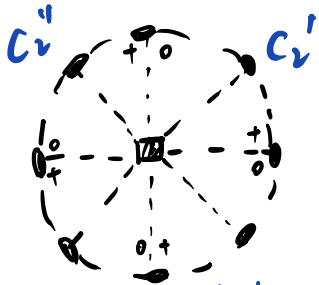
包含一个 n 度主轴 和

n 个与之垂直的 2 度轴

含有 n 度主轴 和

过主轴的 n 个垂直镜面

二. D_4 vs O



C_2'' C_2'

4个 = 度轴，4 为度数

D_4 : C_{42}, C_{x2}, C_{42}^3, E

C_{2x}, C_{2y}, $C_{42}C_{2x} = C_2'(y, x, \bar{z})$

$C_{42}^3 C_{2x} = C_2''(\bar{y}, \bar{x}, z)$

$\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ $\begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$

更细致的计算与比对后面的转化相关

$C_{2x} = \begin{pmatrix} 1 & & \\ & -1 & \\ & & -1 \end{pmatrix}$ $C_{2y} = \begin{pmatrix} -1 & & \\ & 1 & \\ & & -1 \end{pmatrix}$

$C_{42} = \begin{pmatrix} & & -1 & \\ & & & -1 \\ 1 & & & \\ & 1 & & \end{pmatrix}$ $C_{42}^3 = \begin{pmatrix} & & 1 & \\ & & & 1 \\ -1 & & & \\ & -1 & & \end{pmatrix}$

$C_2' = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}$ $C_2'' = \begin{pmatrix} -1 & & & \\ & 1 & & \\ & & -1 & \\ & & & 1 \end{pmatrix}$

$C_{22} = -I_{2\times 2}$ (自成一类)

$E = \begin{pmatrix} 1 & & & \\ & 1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$ $C_{22} = C_{42}^2 = \begin{pmatrix} -1 & & & \\ & -1 & & \\ & & 1 & \\ & & & 1 \end{pmatrix}$

看分块的 2 维表示 $C_2' = \sigma_x$ $C_2'' = -\sigma_x$

$C_{2x} = \sigma_z$, $C_{2y} = -\sigma_z$

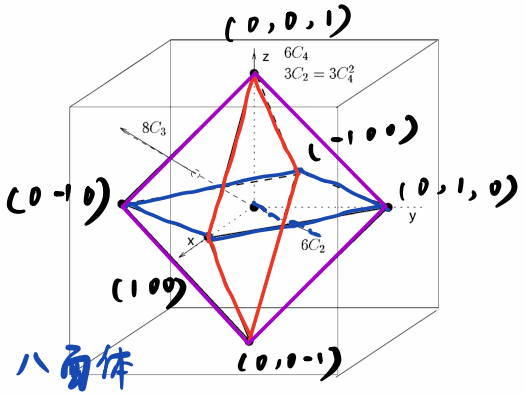
$\sigma_z \sigma_x \sigma_z = -\sigma_x$ (C_2' 同类于 C_2'')

$\sigma_x \sigma_z \sigma_x = -\sigma_z$ (C_{2x} 同类于 C_{2y})

对比 Table 5.4 D_4 群特征标表

C_2' 及 C_2'' $\longrightarrow$ $2C_2''$

C_{2x} 及 C_{2y} $\longrightarrow$ $2C_2'$



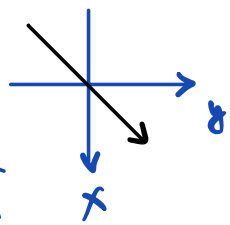
八面体

octahedron inscribed in a cube, illustrating

110 axis $6C_2'$

000 → 110 对称

平行棱边的中点连线



坐标轴的原点在立方体体心

紫 红 蓝 三个正方形 (边长√2)

Table 5.4 Character table for D_4 and the decomposition of the irreducible representations of group O into representations for group D_4

D_4	E	$C_2 = C_4^2$	$2C_4$	$2C_2'$	$2C_2''$	
A_1	1	1	1	1	1	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z, z
B_1	1	1	-1	1	-1	x^2-y^2
B_2	1	1	-1	-1	1	xy
E	2	-2	0	0	0	$\{(xz,yz)$ $(x,y),(R_x,R_y)$

	reducible representations from O group					
E	(2)$_1$	(2)$_2$	(0)$_3$	(2)$_4$	(0)$_5$	$(= A_1 + B_1)$
T_2	3	-1	-1	-1	1	$(= E + B_2)$

O 群

O	E	$8C_3$	$3C_2 = 3C_4^2$ (x, y, z)	$6C_2'$	$6C_4$ (C_{4x}, C_{4x}^3; C_{4y}, C_{4y}^3; C_{4z}, C_{4z}^3)	
Γ_1 A_1	1	1	1	1	1	$(x^2+y^2+z^2)$
Γ_2 A_2	1	1	1	-1	-1	
Γ_{12} E	②$_1$	-1	②$_{2,4}$	⓪$_5$	⓪$_3$	$(x^2-y^2, 3z^2-r^2)$
Γ_{15}' T_1	3	0	-1	-1	1	$\{(x,y,z)$ (R_x, R_y, R_z)
Γ_{25}' T_2	3	0	-1	1	-1	(xy, yz, zx)

D_4 中没有 3 度转动，要对比 D_4 与 O 的特征标表，需要区分 2 度转动的对应关系。

O 群中 C_2' 对应 D_4 群中 C_2''

$3C_2 = 3C_4^2$ (C_{2x}, C_{2y}, C_{2z})

C_{2x}、C_{2y} 相当于 D_4 群中的 C_2'

O 群中的 C_{2z}

即 D_4 群中的 $C_2 = C_4^2$

基函数：（Dresselhaus 的书）

一、对称性操作和基函数

G，R，$\hat{P}_R$：对称操作算符

Γ_n：不可约表示，n 标记表示

可以定义一组 $\{|\Gamma_{nj}\rangle\}$，不可约表示的新矢量 denoted by j　　对一为 component partner

所有的集体构成了矩阵表示 $D^{(\Gamma_n)}(R)$

将 basis vector

处理为 basis function

$$\hat{P}_R |\Gamma_{ni}\rangle = \sum_j D^{(\Gamma_n)}(R)_{ji} |\Gamma_{nj}\rangle$$

$$\langle \Gamma_{nj} | \Gamma_{n'j'} \rangle = \delta_{nn'} \delta_{jj'}$$

$$\int \psi^*_{nj}(\vec{r}) \psi_{n'j'}(\vec{r}) d^3r = \delta_{nn'} \delta_{jj'}$$

$$\int d^3\vec{r} \langle \Gamma_{nj} | \vec{r} \rangle \langle \vec{r} | \Gamma_{n'j'} \rangle = \delta_{nn'} \delta_{jj'}$$

对于 D_3 群：

$$|\Gamma_2, 1\rangle = |x\rangle \quad , \quad |\Gamma_2, 2\rangle = |y\rangle$$

component / partner

$$D^{(\Gamma_2)}(R)_{ja} = \langle \Gamma_{2j} | \hat{P}_R | \Gamma_{2a} \rangle$$

here, D is a clockwise rotation of $2\pi/3$ about z-axis

$D = C_3^{-1}$ ，求表示矩阵

$$D|x\rangle = -\frac{1}{2}|x\rangle - \frac{\sqrt{3}}{2}|y\rangle$$
$$D|y\rangle = \frac{\sqrt{3}}{2}|x\rangle - \frac{1}{2}|y\rangle$$

$$D(D) = \begin{pmatrix} -\frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}$$

将 $|x^2\rangle$ 处理成 $|x\rangle\,|x\rangle$

令 $\hat{x}|x\rangle = x|x\rangle$ ，$\langle x|y\rangle = 0$

Table 4.1. Symmetry operations $\hat{P}_R$ of the group of the equilateral triangle on basis functions taking (x, y, z) into (x', y', z')

$\hat{P}_R$	x'	y'	z'	x'^2	y'^2	z'^2
$E = E$	x	y	z	x^2	y^2	z^2
$C_3 = F$	$\frac{1}{2}(-x + \sqrt{3}y)$	$\frac{1}{2}(-y - \sqrt{3}x)$	z	$\frac{1}{4}(x^2 + 3y^2 - 2\sqrt{3}xy)$	$\frac{1}{4}(y^2 + 3x^2 + 2\sqrt{3}xy)$	z^2
$C_3^{-1} = D$	$\frac{1}{2}(-x - \sqrt{3}y)$	$\frac{1}{2}(-y + \sqrt{3}x)$	z	$\frac{1}{4}(x^2 + 3y^2 + 2\sqrt{3}xy)$	$\frac{1}{4}(y^2 + 3x^2 - 2\sqrt{3}xy)$	z^2
$C_{2(1)} = A$	$-x$	y	$-z$	x^2	y^2	z^2
$C_{2(2)} = B$	$\frac{1}{2}(x - \sqrt{3}y)$	$\frac{1}{2}(-y - \sqrt{3}x)$	$-z$	$\frac{1}{4}(x^2 + 3y^2 - 2\sqrt{3}xy)$	$\frac{1}{4}(y^2 + 3x^2 + 2\sqrt{3}xy)$	z^2
$C_{2(3)} = C$	$\frac{1}{2}(x + \sqrt{3}y)$	$\frac{1}{2}(-y + \sqrt{3}x)$	$-z$	$\frac{1}{4}(x^2 + 3y^2 + 2\sqrt{3}xy)$	$\frac{1}{4}(y^2 + 3x^2 - 2\sqrt{3}xy)$	z^2

有R $f(x,y,z) = f'(x,y,z)$

$\langle z | (E,D,F) | z \rangle = 1$

$\langle z | (A,B,C) | z \rangle = -1$

$\langle R_z | (E,D,F) | R_z \rangle = 1$

$\langle R_z | (A,B,C) | R_z \rangle = -1$

(见下面推导)

test :

$$D(x^2+y^2) = \frac{1}{4}(-x-\sqrt{3}y)^2 + \frac{1}{4}(-y+\sqrt{3}x)^2$$

$$= x^2+y^2$$

$$D(xy) = \frac{1}{2}(-x-\sqrt{3}y) \times \frac{1}{2}(-y+\sqrt{3}x)$$

$$= \frac{1}{4}(-2xy - \sqrt{3}x^2 + \sqrt{3}y^2)$$

$$D(xz) = (-\frac{x}{2} - \frac{\sqrt{3}}{2}y)z$$

$$D(yz) = (-\frac{y}{2} + \frac{\sqrt{3}}{2}x)z$$

.....

$$D(x^2-y^2) = \frac{1}{4}(-x-\sqrt{3}y)^2 - \frac{1}{4}(-y+\sqrt{3}x)^2$$

$$= \frac{1}{4}(-2x^2+2y^2+4\sqrt{3}xy)$$

$$\begin{cases} R(x^2+y^2) = x^2+y^2 \\ R(z^2) = z^2 \end{cases}$$

P_1 : x^2+y^2 , z^2 标量对称变换

P_1' : z , $(x^2+y^2)z$, $R_z = xP_y - yP_x$

↑ ↑

极矢量 轴矢量

polar vector axial vector

pseudo vector

R_z 的计算： Refer to the above table 4.1

$$D(R_z) = \frac{1}{2}(-x-\sqrt{3}y)\times\frac{1}{2}(-p_y+\sqrt{3}p_x) - \frac{1}{2}(-y+\sqrt{3}x)\times\frac{1}{2}(-p_x-\sqrt{3}p_y)$$

$$= R_z$$

$$A(R_z) = (-x)p_y - y(-p_x)$$

$$= -R_z$$

$$B(R_z) = \frac{1}{2}(x-\sqrt{3}y)\frac{1}{2}(-p_y-\sqrt{3}p_x) - \frac{1}{2}(-y-\sqrt{3}x)\frac{1}{2}(p_x-\sqrt{3}p_y)$$

$$= -R_z$$

$$C(R_z) = \frac{1}{2}(x+\sqrt{3}y)\frac{1}{2}(-p_y+\sqrt{3}p_x) - \frac{1}{2}(-y+\sqrt{3}x)\frac{1}{2}(p_x+\sqrt{3}p_y)$$

$$= -R_z$$

$$F(R_z) = \frac{1}{2}(-x+\sqrt{3}y)\frac{1}{2}(-p_y-\sqrt{3}p_x) - \frac{1}{2}(-y-\sqrt{3}x)\frac{1}{2}(-p_x+\sqrt{3}p_y)$$

$$= R_z$$

$$E(R_z) = xp_y - yp_x$$

$$= R_z$$

群论在量子力学中的应用 (part II of dresselhaus's book)

E_n k重简并能级

$\sum_k C_k \psi_{nk}$ 也是 schrödinger 方程的解

$$\hat{R}\psi_{n\alpha} = \sum_j D^{(n)}(R)_{j\alpha}\psi_{nj}$$

不可约表示

$\hat{R}\psi_{n\alpha}$ 的下标 α 是第 α 个简并态中的指标

$D^{(n)}(R)_{j\alpha}$ 矩阵 简并态中的指标

令 $[R, H] = 0$, $[S, H] = 0$ ψ_{nj}

$$\hat{R}\hat{S}\psi_{n\alpha} = \hat{R}\hat{S}\psi_{n\alpha} = \hat{R}\sum_j \psi_{nj} D^{(n)}(S)_{j\alpha}$$

$$= \sum_{jk} \psi_{nk} D^{(n)}(R)_{kj} D^{(n)}(S)_{j\alpha}$$

$$= \sum_k \psi_{nk} [D^{(n)}(R) D^{(n)}(S)]_{k\alpha}$$

$$\therefore D^{(n)}(RS) = D^{(n)}(R) D^{(n)}(S)$$

同样地，$D^{(n)}(R)$ 构成 H 对称群的表示。

不包含负对称性的其它量子数

用态矢 $|\alpha, \Gamma_n, j\rangle$ 标记量子态

高对称的不可约表示 变成 低对称性情形下的可约表示.

能 同等 劈裂的系统（共），不同等 energy levels 的高低

$H = H_0 + H'$

$G \quad G'$ lower symmetry

high symmetry (with fewer symmetry elements)

Group theory tells us exactly what happens to these degeneracies.

Application 1：原子轨道的晶体场劈裂

孤立原子轨道　置于晶体环境中

e.g. D(3)　In quantum mechanics, an atom orbital is a function describing the location and wave-like behavior of an electron in an atom.

一. Background

稀土金属、过渡金属离子　置于晶体场中

(a) 能级劈裂

(b) 劈裂能级的对称性 type

(c) 选择基函数使 H 对角化

二. 晶体中杂质离子的哈密顿量：

第个孤立杂质在晶体场

(第个离子实，多电子)　　(杂质的电子)

$\sum_{ij} \xi_{ij}\, \vec{l}_i \cdot \vec{s}_j$　　i, j: label electron

$\sum_i \gamma_{iM}\, \vec{J}_i \cdot \vec{I}_M$　　M: label impurity

I: nuclear spin

晶体场 V 作为主导的微扰项

特征标是类的函数

- 绕任意轴转动相同角度是一类：
  特征标是转角的函数，

选个最简单的情形吧

$$R(\vec{z},\alpha)(\theta,\varphi)=(\theta,\varphi+\alpha)$$ → 球坐标系

$$R^{-1}(\vec{z},\alpha)(\theta,\varphi)=(\theta,\varphi-\alpha)$$ → Legendre 函数

$$Y_l^m(\theta,\varphi)=P_l^m(\cos\theta)e^{im\varphi}$$

$$P_{\vec{z},\alpha}Y_l^m(\theta,\varphi)=Y_l^m(\theta,\varphi-\alpha)=P_l^m(\cos\theta)e^{im(\varphi-\alpha)}$$

$$=Y_l^m(\theta,\varphi)e^{-im\alpha}$$ 比较（*2）

$$D^l(z,\alpha)_{m'',m}=e^{-im\alpha}\delta_{m',m}$$

(*2)

$$P_{\vec{w},\alpha}Y_l^m(\theta,\varphi)=\sum_{m'}D^l(\vec{w},\alpha)_{m'm}Y_l^{m'}(\theta,\varphi)$$

表示矩阵：对角阵

$$D^l(z,\alpha)=\begin{bmatrix} e^{-il\alpha} & & & 0 \\ & e^{-i(l-1)\alpha} & 0 & \\ & 0 & \ddots & \\ 0 & & & e^{il\alpha} \end{bmatrix}$$

所以，在第 l 个表示中，转角为 α 这一类的特征标为

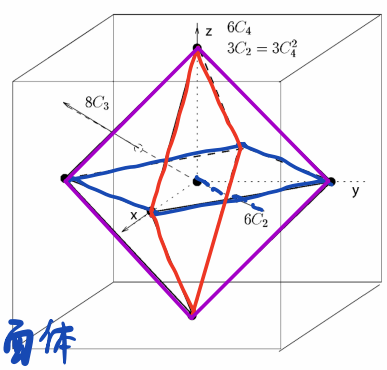
$$\chi^l(\alpha)=\sum_{m=-l}^{l}e^{-im\alpha}=e^{-il\alpha}\sum_{n=0}^{2l}(e^{i\alpha})^n$$

$$=e^{-il\alpha}\frac{e^{i(2l+1)\alpha}-1}{e^{i\alpha}-1}=\frac{\sin(l+1/2)\alpha}{\sin(\alpha/2)}$$

要处理的问题：

Cubic Crystal Field Environment for a paramagnetic transition metal ion

e.g an iron impurity in a cubic host crystal. Assume further that this impurity goes into a substitutional lattice site, and is surrounded by a regular octahedron of negative ions.



八面体

octahedron inscribed in a cube, illustrating

110 axis

000 → 110 射线

$6C_2'$

平行棱边和中点连线

check $\Gamma_{l=1}$ 的特征标。 (Table 5.1)

$$\chi^{(l=1)}(\alpha) = \frac{\sin(\frac{3}{2}\alpha)}{\sin(\frac{\alpha}{2})}$$

表格中，$\Gamma_{l=1}$ 的 T_1 与 O 群的 T_1 不一样

C_2 和 C_2' 一样. 所以 Γ_l 中的 T_1 是不对的。

$$\chi^{(l=1)}(\pi) = \frac{\sin(\frac{3}{2}\pi)}{\sin(\frac{\pi}{2})} = -1$$

$$C_3: \chi^{(l=1)}(\tfrac{2\pi}{3}) = \frac{\sin(\frac{3}{2}\times\frac{2\pi}{3})}{\sin(\frac{1}{2}\times\frac{2\pi}{3})} = 0$$

$$C_4: \chi^{(l=1)}(\tfrac{\pi}{2}) = \frac{\sin(\frac{3}{2}\times\frac{\pi}{2})}{\sin(\frac{\pi}{4})} = 1$$

这样，O 群中的 T_1 对应的特征标正确。

三. 顺磁过渡金属离子的立方晶体场劈裂

孤立 $l=2$ (d 轨道) 的磁性杂质原子

$$\chi^{l}(\alpha) = \frac{\sin\left(l+\frac{1}{2}\right)\alpha}{\sin\frac{1}{2}\alpha}$$

(参看完全转动群笔记关于 SO(3) 章节.)

$\chi^{(l)}(E) = 5$

$\chi^{(l)}(C_3) = -1$

$\chi^{(l)}(C_2) = 1$

$\chi^{(l)}(C_4) = -1$

86 5 Splitting of Atomic Orbitals in a Crystal Potential

Table 5.1. Character table for O and decomposition of the angular momenta representations into the irreducible representations of group O

	O	E	$8C_3$	$3C_2 = 3C_4^2$	$6C_2'$	$6C_4$
Γ_1	A_1	1	1	1	1	1
Γ_2	A_2	1	1	1	−1	−1
Γ_{12}	E	2	−1	2	0	0
$\Gamma_{15'}$	T_1	3	0	−1	−1	1
$\Gamma_{25'}$	T_2	3	0	−1	1	−1
$\Gamma_{\ell=0}$	A_1	1	1	1	1	1
$\Gamma_{\ell=1}$	T_1	3	0	−1	-1	+1
$\Gamma_{\ell=2}$	$E + T_2$	5	−1	1	1	−1
$\Gamma_{\ell=3}$	$A_2 + T_1 + T_2$	7	1	−1	−1	−1
$\Gamma_{\ell=4}$	$A_1 + E + T_1 + T_2$	9	0	1	1	1
$\Gamma_{\ell=5}$	$E + 2T_1 + T_2$	11	−1	−1	−1	1

refer 右 Table 5.1 P86. dressed bands

可约表示分解公式

$$\chi^{\text{reducible}}(C_k) = \sum_j a_j \chi^{(\Gamma_j)}(C_k)$$

$$a_j = \frac{1}{h}\sum_k N_k \chi^{(\Gamma_j)}(C_k)^* \chi^{\text{reducible}}(C_k)$$

h: group order; N_k: # of elements in class k

SO(3) 对 O 群的分解

E	$8C_3$	$3C_2 = 3C_4^2$	$6C_2'$	$6C_4$

$$a_{A_1} = \frac{1}{24}[1\times1\times5 + 8\times1\times(-1) + 3\times1\times1 + 6\times1\times1 + 6\times1\times(-1)]$$

$$= 0$$

$$a_{A_2} = \frac{1}{24}[1\times1\times5 + 8\times1\times(-1) + 3\times1\times1 + 6\times(-1)\times1 + 6\times(-1)\times(-1)]$$

$$= 0$$

$$a_E = \frac{1}{24}[1\times2\times5 + 8\times(-1)\times(-1) + 3\times2\times1 + 6\times0\times1 + 6\times0\times(-1)]$$

$$= 1$$

$$a_{T_1} = 0$$

$$a_{T_2} = 1$$

$\Gamma_{rot}^{(2)} = E^{(dim:2)} \oplus T_2^{(dim:3)}$

↓

$l=2$ 的完全转动群的不可约表示

single impurity

eq. (5.1) $\sum_i$

↓ 等离质的多电子 label 电子

加入 impurity 后增加的 interaction term

所生成基态到的 D 群对称性

(真空 → 晶体物)

O $\xrightarrow[\text{Symmetry}]{\text{apply stress}}$ (D_4 Symmetry)

(见上图和下面的特征标表)

from table 5.1 → 5.4

(O group) (D4)

E: $a_{A_1} = \frac{1}{8}[1\times1\times2 + 1\times1\times2 + 2\times1\times0 + 2\times1\times2 + 2\times1\times0] = 1$

$a_{B_1} = 1$ else $= 0$

$$E = A_1 + B_1$$

T_2: $a_{A_1} = \frac{1}{8}[1\times1\times3 + 1\times1\times(-1) + 2\times1\times(-1) + 2\times1\times(-1) + 2\times1\times1] = 0$

$a_{A_2} = 0$

$a_{B_1} = 0$

$a_{B_2} = 1$

$a_E = 1$

$$T_2 = B_2 + E$$

$$\Gamma^{(1)}_{104} = A_1 + B_1 + B_2 + E$$

Table 5.4 Character table for D_4 and the decomposition of the irreducible representations of group O into representations for group D_4

D_4	E	$C_2 = C_4^2$	$2C_4$	$2C_2'$	$2C_2''$	
A_1	1	1	1	1	1	x^2+y^2, z^2
A_2	1	1	1	-1	-1	R_z, z
B_1	1	1	-1	1	-1	x^2-y^2
B_2	1	1	-1	-1	1	xy
E	2	-2	0	0	0	$\{(xz, yz)$ $(x,y), (R_x, R_y)$

	reducible	representations	from	O	group	
E	2	2	0	2	0	$(\equiv A_1 + B_1)$
T_2	3	-1	-1	-1	1	$(\equiv E + B_2)$

Table 5.5 Decomposition of the $l=2$ angular momentum level into the irreducible representations of group D_4

	E	C_2	$2C_4$	$2C_2'$	$2C_2''$	
$P_{tot}^{(2)}$	5	1	-1	1	1	$A_1 + B_1 + B_2 + E$

不可约表示的变换性质

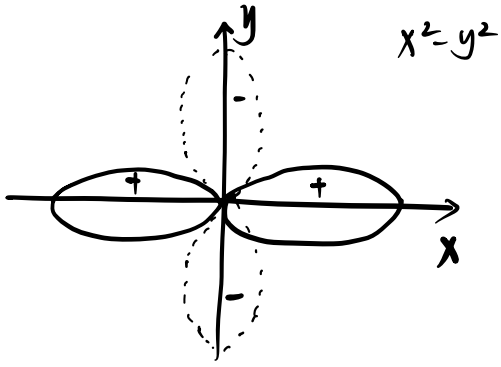
即基函数的变换性质

O 群

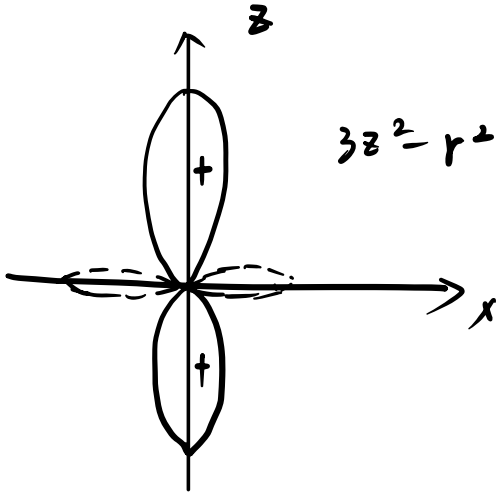
O	E	$8C_3$	$3C_2 = 3C_4^2$	$6C_2'$	$6C_4$	
Γ_1 A_1	1	1	1	1	1	$(x^2+y^2+z^2)$
Γ_2 A_2	1	1	1	-1	-1	
Γ_{12} E	2	-1	2	0	0	$(x^2-y^2,\ 3z^2-r^2)$
Γ_{15}' T_1	3	0	-1	-1	1	$\begin{cases}(x,y,z)\\(R_x,R_y,R_z)\end{cases}$
Γ_{25}' T_2	3	0	-1	1	-1	$(xy,\ yz,\ zx)$

Consider the case of a d-electron in a a cubic field : $x=\pm a,\ y=\pm a,\ z=\pm a$ (host ions)

d 轨道波函数，



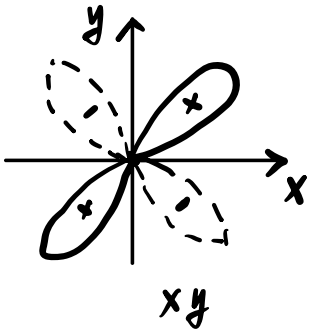
x^2-y^2



$3z^2-r^2$

波函数变量更多
与近邻

相比于 四叶草 轨道波函数
计 吸引相互作用 : 能量更低
排斥 ~ : ~ 高



$(x^2-y^2,\ 3z^2-r^2)$: E

↙ A_1 ↘ B_1

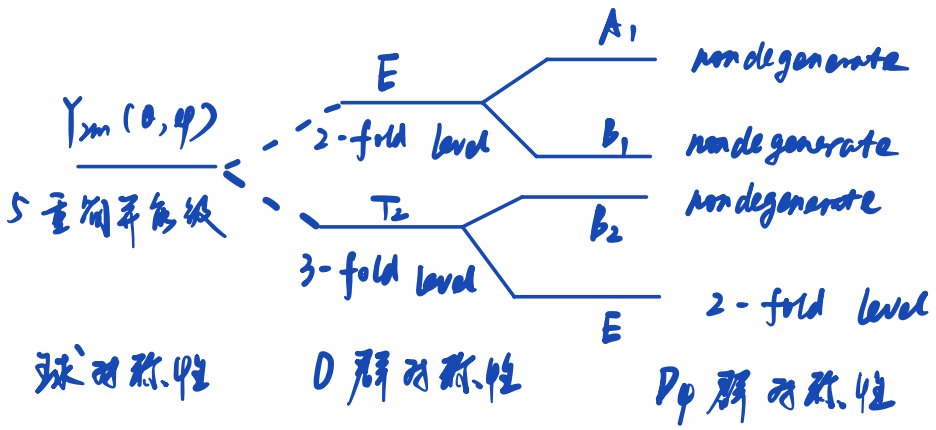
$x^2+y^2,\ z^2$ x^2-y^2

$E = A_1 + B_1$

(0) (Dq)

Fig.5.4

d-level splitting in octahedral and Dq crystal fields



A_{1g} g: even parity

A_{1u} u: odd ~

根据晶体场 挑选为相应的球谐函数

拥有 A_{1g} 对称性

晶体场 拥有 O_h 变换下的不变性

最小 non-trivial 轨道电子数 $l=4$

Table 5.6. Splitting of angular momentum in cubic symmetry O_h

ℓ	A_{1g}	A_{2g}	E_g	T_{1g}	T_{2g}	A_{1u}	A_{2u}	E_u	T_{1u}	T_{2u}
0	1									
1									1	
2			1		1					
3							1		1	1
4	1		1	1	1					
5								1	2	1
6	1	1	1	1	2					
7							1	1	2	2
8	1		2	2	2					
9						1	1	1	3	2
10	1	1	2	2	3					
11							1	2	3	3
12	2	1	2	3	3					
13						1	1	2	4	3
14	1	1	3	3	4					
15						1	2	2	4	4

出现两次

(球对称基矢)

$l=0$ >> O_h 对称性

说明：$l=4$ 遵循 O_h 对性性的 A_{1g} 表示的变换性质

(totally symmetric irreducible reps)

$$V_{cubic} = \frac{35e}{4a^5}\left[(x^4+y^4+z^4) - \frac{3}{5}r^4\right]$$

$x=\pm a$, $y=\pm a$, $z=\pm a$ 处的晶体离子

提供给原来自由原子(离子)的微扰晶体势场

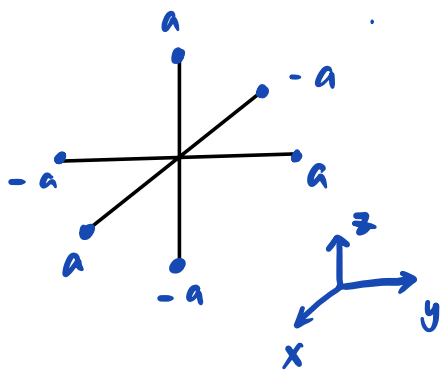
$Y_{lm}(\theta,\phi)$, $Y_{4,0}$, $Y_{4,4}$, $Y_{4,-4}$

(见网站角级劈裂笔记)

Point sources:

$\nabla^2 V = 0$

$r^l Y_{lm}(\theta,\phi)$



$Y_{4,0}$: z^4

$Y_{4,\pm4}$: (x^4+y^4)

We can check the predictions from group theory by obtaining the crystal field analytically.

The contribution from an ion at $x=-a$ at the field point $\vec{r}$ denoted by (x, y, z) is

$$V_{x=-a} = \frac{e}{|\vec{r}|} = \frac{e}{a\sqrt{(1+x/a)^2+(y/a)^2+(z/a)^2}}$$

where e is the charge e on the electron and ε is a small dimensionless quantity if considering (x, y, z) in the neighborhood of the origin 0. (impuring ion position) Then using the binomial expansion: (二项式展开)

$$(1+\varepsilon)^{-1/2} = 1 - \frac{1}{2}\varepsilon + \frac{3}{8}\varepsilon^2 - \frac{5}{16}\varepsilon^3 + \frac{35}{128}\varepsilon^4 + \cdots$$

sum up: $V_{x=-a} + V_{x=a} + V(x\to y) + V(x\to z)$

So that the perturbation that will lift the degeneracy of the free atom is of the form:

$$V_{cubic} = \frac{35e}{4a^5}\left[(x^4+y^4+z^4) - \frac{3}{5}r^4\right]$$